

25/07/26

COL - ASSESSMENT TOOL - I

① Class Hierarchy :

CLASS Person

DATA

Name

Age

METHOD getPersonDetails()

READ Name

READ Age

END METHOD

END CLASS

CLASS student INHERITS Person

DATA

Register Number

marks[5]

Average

METHOD calculateAverage()

sum $\leftarrow$ 0FOR i $\leftarrow$ 0 TO 4sum $\leftarrow$ sum + marks[i]

END FOR

Average $\leftarrow$ sum/5

END METHOD

END CLASS

CLASS ResearchStudent INHERITS student

DATA

ResearchArea

METHOD displayDetails()

PRINT Name

PRINT Age

PRINT RegisterNumber

PRINT Average

PRINT ResearchArea

END METHOD

END CLASS

BEGIN

CREATE ResearchStudent R

R.getPersonDetails()

READ R.RegisterNumber

FOR $i \leftarrow 0$ TO 4

READ R.Marks[i]

END FOR

READ R.ResearchArea

R.calculateAverage()

R.displayDetails()

END

② REMOVE DUPLICATES ;

Complete the missing pseudocode

BEGIN

READ N

READ ARRAY A

FOR $i \leftarrow 0$ TO $N-1$

FOR $j \leftarrow i+1$ TO $N-1$

IF $A[i] == A[j]$ THEN

DELETE $A[j]$

$N \leftarrow N-1$

$j \leftarrow j-1$

END IF

END FOR

END FOR

PRINT Modified Array

END

- ③ Area using overloading
Debug the Pseudocode

Incorrect Lines

RETURN length * length

RETURN length * breadth

Correct program:-

CLASS Area

METHOD area(length)

RETURN length * length

END METHOD

METHOD area(length, breadth)

RETURN length * breadth

END METHOD

END CLASS

- ④ Trace the Execution

Sum $\leftarrow$ 0

FOR $i \leftarrow 1$ TO 5

IF $i \text{ MOD } 2 == 0$

Sum $\leftarrow$ Sum + i

ELSE

Sum $\leftarrow$ Sum - i

END IF

END FOR

PRINT Sum

Trace Table

i	operation	sum
1	0-1	-1
2	-1+2	1
3	1-3	-2
4	-2+4	2
5	2-5	-3

output i: -3

⑤ Second Largest

Write Efficient Pseudocode

BEGIN

 READ N

 READ ARRAY A

 largest $\leftarrow -\infty$

 second $\leftarrow -\infty$

 FOR $i \leftarrow 0$ TO $N-1$

 IF $A[i] > \text{largest}$ THEN

 second $\leftarrow \text{largest}$

 largest $\leftarrow A[i]$

 ELSE IF $A[i] > \text{second}$ AND $A[i] \neq \text{largest}$

 THEN

 second $\leftarrow A[i]$

 END IF

 END FOR

 PRINT second

END

⑥ Payment System

Interface:

```
INTERFACE Payment
```

```
    METHOD pay(amount)
```

```
END INTERFACE
```

```
CLASS CreditCard IMPLEMENTS Payment
```

```
    METHOD pay(amount)
```

```
        PRINT "Paid using Credit Card :", amount
```

```
    END METHOD
```

```
END CLASS
```

```
CLASS UPI IMPLEMENTS Payment
```

```
    METHOD pay(amount)
```

```
        PRINT "Paid using UPI :", amount
```

```
    END METHOD
```

```
END CLASS
```

```
CLASS NetBanking IMPLEMENTS Payment
```

```
    METHOD pay(amount)
```

```
        PRINT "Paid using Net Banking :", amount
```

```
    END METHOD
```

```
END CLASS
```

```
BEGIN
```

```
    READ choice
```

```
    READ amount
```

```
    IF choice = 1
```

```
        obj ← CreditCard
```

```
    ELSE IF choice = 2
```

```
        obj ← UPI
```

```
    ELSE
```

```
        obj ← NetBanking
```

END IF

Obj.pay (amount)

END

⑦ Runtime Polymorphism ;

Complete the pseudocode

CLASS Animal

METHOD speak()

PRINT "Animal"

END METHOD

END CLASS

CLASS Dog INHERITS Animal

METHOD speak()

PRINT "Barks"

END METHOD

END CLASS

Animal obj ← NEW Dog()

obj ← speak()

Output : Barks.

⑧ Constructor ;

Finding Logical Errors

CLASS Employee

DATA

id

name

constructor (id, name)

THIS.id ← id

THIS.name ← name

END CONSTRUCTOR

END CLASS

⑨

Vehicle Hierarchy;

Case study

CLASS vehicle

DATA

vehicleNo

ownerName

METHOD display()

PRINT vehicleNo

PRINT ownerName

END METHOD

END CLASS

CLASS car INHERITS vehicle

DATA

FuelType

METHOD display()

PRINT vehicleNo

PRINT ownerName

PRINT FuelType

END METHOD

END CLASS

CLASS truck INHERITS vehicle

DATA

loadCapacity

METHOD display()

Print vehicleNo

Print ownername

Print loadCapacity

END METHOD

END CLASS

BEGIN

READ N

FOR $i \leftarrow 1$ TO N

 READ type

 IF type = "car" Then

 Create Car C

 READ C.vehicleNo

 READ C.ownerName

 READ C.fuelType

 STORE C

 ELSE

 CREATE Truck T

 READ T.vehicleNo

 READ T.ownerName

 READ T.loadCapacity

 STORE T

 END IF

END FOR

FOR each vehicle v

 v.display()

END FOR

END.